

Recommendation System

A **Recommendation System** is a machine learning system that suggests items to users based on their interests, behavior, or preferences.

It helps users find:

- Movies on Netflix
 - Videos on YouTube
 - Products on Amazon
 - Songs on Spotify
 - Friends on Instagram/Facebook
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Simple Definition

A recommendation system predicts:

“What item will the user like next?”

Real-Life Examples

- **Netflix** → recommends movies/web series
 - **Amazon** → recommends products
 - **YouTube** → recommends videos
 - **Spotify** → recommends songs
 - **Instagram** → recommends reels/posts
-

Why Recommendation Systems Are Important

They help companies:

- Increase user engagement
 - Improve user experience
 - Increase sales and watch time
 - Keep users active on the platform
-

Types of Recommendation Systems

1. Content-Based Filtering

This system recommends items similar to what the user already likes.

Example

If you watch:

- Action movies
- Superhero movies

Then the system recommends:

- Similar action/superhero movies

How It Works

It checks:

- Keywords
- Categories
- Tags
- Features of items

Example Table

User Likes	Recommended
Cricket videos	More cricket videos
Horror movies	Similar horror movies

Advantages

- Personalized recommendations
- No need for other users' data

Disadvantages

- Limited variety
- Recommends very similar items only

2. Collaborative Filtering

This system recommends items based on similar users.

Idea

“Users who liked the same things as you may also like this item.”

Example

- User A likes:
 - Java
 - Python
- User B likes:
 - Java
 - Python
 - Node.js

Then Node.js may be recommended to User A.

Types of Collaborative Filtering

A. User-Based Collaborative Filtering

Find users with similar interests.

B. Item-Based Collaborative Filtering

Find similar items.

Example:

- Users who bought iPhone also bought AirPods.
-

Recommendation Matrix Example

User **Movie A** **Movie B** **Movie C**

A 5 4 ?

B 5 4 5

System predicts:

- User A may also like Movie C.

3. Hybrid Recommendation System

Combination of:

- Content-Based Filtering
- Collaborative Filtering

Used by:

- Netflix
- Amazon
- YouTube

Because hybrid systems are more accurate.

Machine Learning Algorithms Used

Some common algorithms:

- KNN
 - Matrix Factorization
 - Decision Trees
 - Deep Learning
 - Neural Networks
 - Clustering Algorithms
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Collaborative Filtering

Collaborative Filtering is a recommendation technique that suggests items to a user based on the preferences and behavior of other similar users.

It works on the idea:

“People with similar interests usually like similar things.”

Simple Example

Suppose:

User Likes

A Java, Python

B Java, Python, Node.js

Since both users like Java and Python, the system predicts:

→ User A may also like Node.js

This is Collaborative Filtering.

Real-Life Examples

- **Netflix → movie recommendations**
 - **Amazon → product suggestions**
 - **Spotify → music recommendations**
 - **YouTube → video recommendations**
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Main Idea

Collaborative Filtering uses:

- **User behavior**
- **Ratings**
- **Likes**
- **Purchases**
- **Watch history**

instead of item features.

Types of Collaborative Filtering

1. User-Based Collaborative Filtering

Find users similar to the current user.

Workflow

Find Similar Users



Check What They Like



Recommend Those Items

Example

User Movies Liked

A Avengers, Batman

B Avengers, Batman, Superman

System recommends:

→ Superman to User A

2. Item-Based Collaborative Filtering

Find similar items based on user interactions.

Workflow

Find Similar Items



Recommend Related Items

Example

If many users buy:

- **iPhone**
- **AirPods**

Then the system learns:

iPhone and AirPods are related.

So when someone buys an iPhone, it recommends AirPods.

User-Item Matrix

Collaborative filtering often uses a matrix like this:

User Movie A Movie B Movie C

A 5 4 ?

B 5 4 5

C 2 1 1

The system predicts:

→ User A may rate Movie C highly.

How Similarity is Calculated

Algorithms measure similarity using:

- Cosine Similarity
- Pearson Correlation
- Euclidean Distance

Example:

If two users give similar ratings, they are considered similar.

Advantages

- No need to know item details
 - Good personalized recommendations
 - Improves user experience
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Disadvantages

1. New users/items have no data.
2. Most users rate only a few items.
3. Large systems need high computation.

Content-Based Filtering

Content-Based Filtering is a recommendation technique that suggests items similar to what a user already likes.

It focuses on:

- **Item features**
 - **Categories**
 - **Keywords**
 - **Tags**
 - **User preferences**
-

Simple Idea

“If you liked this item before, you may like similar items again.”

Real-Life Example

Suppose you watch:

- **Action movies**
- **Superhero movies**

Then Netflix recommends:

- **More action/superhero movies**

because their content is similar.

How It Works

The system:

1. **Collects user interests**
 2. **Analyzes item features**
 3. **Finds similar items**
 4. **Recommends them**
-

Workflow

User Likes an Item



Extract Item Features



Find Similar Items



Recommend Similar Content

Example

User Watched:

- **Iron Man**
- **Avengers**

Features:

- **Action**
- **Marvel**
- **Superhero**
- **Sci-fi**

Recommended:

- **Captain America**
- **Thor**
- **Doctor Strange**

because they share similar features.

Where It Is Used

- **Netflix → movie recommendation**
- **YouTube → similar videos**
- **Spotify → similar songs**
- **News apps → related articles**
- **E-commerce → related products**

User Profile

The system creates a user profile based on interests.

Example:

User Interest	Weight
---------------	--------

Action Movies	High
---------------	------

Comedy	Medium
--------	--------

Horror	Low
--------	-----

Then recommendations are generated using this profile.

Feature Vector

Items are converted into numerical values called feature vectors.

Example:

Movie	Action	Comedy	Sci-Fi
Avengers	1	0	1
Thor	1	0	1
Comedy Movie	0	1	0

The system compares vectors to find similarity.

Advantages

1. Recommendations depend on user interests.
 2. Works independently.
 3. Can recommend newly added items easily.
-

Disadvantages

1. Suggests only similar items.

Example:

- If you watch only action movies,
- you may never discover comedy movies.

2. System must understand item properties correctly.

3. Recommendations become repetitive.

Artificial Neural Network (ANN)

An Artificial Neural Network (ANN) is a machine learning model inspired by the human brain.

It is made of interconnected nodes called neurons, which work together to learn patterns from data.

ANN is mainly used in:

- **Image recognition**
 - **Speech recognition**
 - **Recommendation systems**
 - **Chatbots**
 - **Medical diagnosis**
 - **Self-driving cars**
-

Structure of ANN

ANN mainly has 3 layers:

Input Layer → Hidden Layer → Output Layer

1. Input Layer

Receives input data.

Example:

- **Age**
 - **Salary**
 - **Marks**
 - **Pixels of an image**
-

2. Hidden Layer

Performs calculations and pattern learning.

This is where:

- **weights**

- activation functions
- learning

happen.

There can be multiple hidden layers.

3. Output Layer

Produces the final result.

Example:

- Spam / Not Spam
 - Cat / Dog
 - Price prediction
-

Basic Working of ANN

Input Data

↓

Multiply by Weights

↓

Apply Activation Function

↓

Generate Output

↓

Adjust Weights using Learning

Artificial Neuron

A neuron receives:

- Inputs
- Weights
- Bias

Then calculates output.

The basic neuron equation is:

$$y = f(\sum wx + b)$$

Where:

- x = input
 - w = weight
 - b = bias
 - f = activation function
 - y = output
-

Important Terms

1. Weights

Weights decide:

How important an input is.

Higher weight = higher importance.

2. Bias

Bias helps shift the output.

It improves flexibility of the model.

3. Activation Function

Determines whether a neuron should activate.

Common activation functions:

- Sigmoid
 - ReLU
 - Tanh
 - Softmax
-

Example of ANN

Suppose we want to identify:

- Cat or Dog image

Input:

- Image pixels

ANN learns:

- Ear shape
- Eyes
- Fur pattern

Output:

- Cat
 - Dog
-

Training Process

ANN learns using training data.

Steps:

Step 1: Forward Propagation

Input moves from input layer → output layer.

Step 2: Error Calculation

Compare predicted output with actual output.

Step 3: Backpropagation

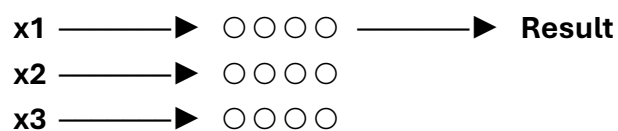
Adjust weights to reduce error.

Step 4: Repeat

Model improves gradually.

Simple ANN Diagram

Input Layer Hidden Layer Output Layer



Applications of ANN

1. Image Recognition

- **Face detection**
- **Object detection**

2. Speech Recognition

- **Siri**
- **Google Assistant**

3. Recommendation Systems

- **Netflix**
- **YouTube**

4. Medical Field

- **Disease prediction**

5. Finance

- **Fraud detection**
- **Stock prediction**

Advantages of ANN

- **Learns complex patterns**
- **High accuracy**
- **Works with large data**
- **Automatic feature learning**

Disadvantages of ANN

- **Requires large data**
- **Training takes time**
- **High computational power**
- **Difficult to interpret**

Perceptron

A Perceptron is the simplest type of Artificial Neural Network (ANN).

It is a single artificial neuron used for:

- **Binary classification**
- **Decision making**

It gives output in two classes:

- **Yes / No**
 - **0 / 1**
 - **True / False**
-

Simple Definition

A Perceptron is a single-layer neural network that takes inputs, applies weights, and produces a binary output.

Invented By

The perceptron was introduced by:

Frank Rosenblatt

in 1958.

Structure of Perceptron

A perceptron contains:

- 1. Input values**
 - 2. Weights**
 - 3. Bias**
 - 4. Activation function**
 - 5. Output**
-

Basic Diagram

x1 —►
x2 —► [Perceptron] —► Output
x3 —►

Working of Perceptron

Steps:

Input Data

↓

Multiply by Weights

↓

Add Bias

↓

Apply Activation Function

↓

Generate Output

Mathematical Formula

The perceptron calculates:

$$y = f(\sum w_i x_i + b)$$

Where:

- x_i = input values
 - w_i = weights
 - b = bias
 - f = activation function
 - y = output
-

Step-by-Step Example

Suppose:

- $x_1 = 2$
- $x_2 = 3$

Weights:

- $w_1 = 1$
- $w_2 = 2$

Bias:

- $b = 1$

Calculation:

$$z = (2 \times 1) + (3 \times 2) + 1 = 9$$

Now apply activation function.

If:

- $z > 0 \rightarrow \text{Output} = 1$
- Otherwise $\rightarrow \text{Output} = 0$

So output = 1.

Activation Function

The perceptron mostly uses the Step Function.

$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

This means:

- Positive value $\rightarrow 1$
 - Negative value $\rightarrow 0$
-

Types of Perceptron

1. Single Layer Perceptron (SLP)

- Only one layer
- Solves simple linear problems

Example:

- AND gate
 - OR gate
-

2. Multi Layer Perceptron (MLP)

- Multiple hidden layers
- Solves complex problems

Used in:

- Deep learning
 - Image recognition
 - NLP
-

Perceptron Learning Algorithm

The perceptron learns by updating weights.

Steps:

1. Take input
 2. Predict output
 3. Compare with actual output
 4. Adjust weights if wrong
 5. Repeat until accurate
-

Weight Update Formula

$$w_{new} = w_{old} + \eta(t - y)x$$

Where:

- η = learning rate
 - t = target output
 - y = predicted output
 - x = input
-

Example of Perceptron in Real Life

Spam Detection

Input:

- Contains “Free”

- **Contains “Offer”**
- **Contains links**

Output:

- **Spam = 1**
 - **Not Spam = 0**
-

Advantages

- **Simple and fast**
 - **Easy to implement**
 - **Good for binary classification**
-

Disadvantages

Cannot Solve Non-Linear Problems

Perceptron fails for:

- **XOR gate**
- **Complex patterns**

Because it only handles linearly separable data.

Multilayer Network

A Multilayer Network is an Artificial Neural Network (ANN) that contains:

- One input layer
- One or more hidden layers
- One output layer

It is also called:

Multilayer Perceptron (MLP)

because it contains multiple perceptrons connected together.

Simple Definition

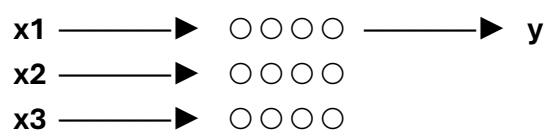
A Multilayer Network is a neural network with multiple layers of neurons that can learn complex patterns and solve non-linear problems.

Structure of Multilayer Network

Input Layer → Hidden Layer(s) → Output Layer

Diagram

Input Layer Hidden Layer Output Layer



Layers in Multilayer Network

1. Input Layer

Receives input data.

Example:

- Age
- Salary

- Image pixels
 - Sensor values
-

2. Hidden Layer

Performs calculations and learns patterns.

This layer:

- Extracts features
- Detects relationships
- Improves learning

A network can have:

- One hidden layer
- Multiple hidden layers

More hidden layers = deeper learning.

3. Output Layer

Produces the final result.

Examples:

- Cat / Dog
 - Spam / Not Spam
 - Price prediction
-

Working of Multilayer Network

Input Data



Forward Propagation



Hidden Layer Processing



Output Prediction



Error Calculation

↓

Backpropagation

↓

Weight Update

Mathematical Representation

Each neuron calculates:

$$y = f(\sum w_i x_i + b)$$

Where:

- x_i = inputs
 - w_i = weights
 - b = bias
 - f = activation function
-

Why Hidden Layers Are Important

Single-layer perceptrons can solve only:

- Linear problems

But multilayer networks solve:

- Non-linear problems
- Complex pattern recognition

Example:

- XOR gate
 - Face recognition
 - Speech recognition
-

Activation Functions Used

Common activation functions:

1. Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Output range:

- 0 to 1
-

2. ReLU (Rectified Linear Unit)

$$f(x) = \max(0, x)$$

Most commonly used in deep learning.

3. Tanh

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Output range:

- -1 to 1
-

Forward Propagation

Data moves:

- Input layer → hidden layers → output layer

Each neuron:

1. Receives input
 2. Applies weights
 3. Uses activation function
 4. Sends output forward
-

Backpropagation

Backpropagation is used to:

- Reduce errors
- Improve predictions

Steps:

1. Compare predicted output with actual output
 2. Calculate error
 3. Adjust weights
 4. Repeat training
-

Applications of Multilayer Network

1. Image Recognition

- Face detection
- Object recognition

2. Natural Language Processing

- Chatbots
- Translation

3. Recommendation Systems

- Netflix
- YouTube

4. Medical Diagnosis

- Disease prediction

5. Self-Driving Cars

- Traffic sign detection
-

Advantages

- Solves complex problems
 - Learns non-linear relationships
 - High prediction accuracy
 - Automatic feature learning
-

Disadvantages

- Requires large training data

- **Training can be slow**
- **Needs high computational power**
- **Difficult to understand internally**

Backpropagation

Backpropagation is a training algorithm used in Artificial Neural Networks (ANNs) to reduce errors and improve prediction accuracy.

It works by:

- 1. Calculating the error**
 - 2. Sending the error backward**
 - 3. Updating weights**
-

Simple Definition

Backpropagation is a learning process in neural networks where errors are propagated backward to adjust weights and improve predictions.

Why Backpropagation is Needed

Neural networks initially make wrong predictions because weights are random.

Backpropagation helps the network:

- Learn from mistakes**
 - Reduce error**
 - Improve accuracy**
-

Main Idea

Prediction

↓

Calculate Error

↓

Send Error Backward

↓

Update Weights

↓

Improve Prediction

Working of Backpropagation

Backpropagation has two major phases:

1. Forward Propagation

Input moves:

- **Input layer → hidden layer → output layer**

The network generates prediction.

2. Backward Propagation

Error is calculated and propagated backward.

Weights are updated to minimize error.

Step-by-Step Process

Step 1: Input Data

Example:

- **Image**
- **Text**
- **Numbers**

Input enters the network.

Step 2: Forward Pass

Each neuron calculates output:

$$y = f(\sum w_i x_i + b)$$

Prediction is generated.

Step 3: Calculate Error

Compare:

- **Actual output**
- **Predicted output**

Common error function:

$$E = \frac{1}{2} (Target - Output)^2$$

If error is high:

- network must improve.
-

Step 4: Backward Pass

Error travels backward:

- Output layer → hidden layer → input layer

The network calculates:

- Which weights caused error
-

Step 5: Update Weights

Weights are adjusted using gradient descent.

Weight update formula:

$$w_{new} = w_{old} - \eta \frac{\partial E}{\partial w}$$

Where:

- η = learning rate
 - E = error
 - w = weight
-

Learning Rate

Learning rate controls:

How much weights should change.

- Small learning rate → slow learning
 - Large learning rate → unstable learning
-

Role of Activation Function

Activation functions help networks learn non-linear patterns.

Common functions:

- Sigmoid
- ReLU
- Tanh

Example sigmoid:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Simple Example

Suppose:

- Network predicts: 70
- Actual answer: 100

Error:

- 30

Backpropagation:

- finds incorrect weights
 - adjusts them
 - improves next prediction
-

Diagram

Forward Pass:

Input → Hidden Layer → Output

Backward Pass:

Error ← Hidden Layer ← Output

Advantages

- Improves model accuracy
- Learns complex patterns
- Works well in deep learning

- **Efficient training method**
-

Disadvantages

- **Training can be slow**
 - **Requires large data**
 - **May get stuck in local minima**
 - **Needs computational power**
-

Applications

1. Image Recognition

- **Face detection**
- **Object classification**

2. Speech Recognition

- **Voice assistants**

3. NLP

- **Chatbots**
- **Translation**

4. Recommendation Systems

- **Netflix**
- **YouTube**

Introduction to Deep Learning

Deep Learning is a subset of Machine Learning that uses deep neural networks with many hidden layers to learn complex patterns from large amounts of data.

It is inspired by the working of the human brain.

Simple Definition

Deep Learning is an advanced form of machine learning that uses multilayer neural networks to automatically learn features and make intelligent decisions.

Why It Is Called “Deep”

In neural networks:

- Few hidden layers → Normal ANN
- Many hidden layers → Deep Learning

“Deep” means:

Multiple hidden layers between input and output.

Basic Structure

Input Layer

↓

Hidden Layer 1

↓

Hidden Layer 2

↓

Hidden Layer 3

↓

Output Layer

Deep Learning vs Machine Learning

Machine Learning

Deep Learning

Requires feature extraction manually Learns features automatically

Machine Learning

Works well with small data

Simpler models

Faster training

Deep Learning

Needs large data

Complex neural networks

Slower training

Working of Deep Learning

Step 1: Input Data

Input can be:

- **Images**
 - **Audio**
 - **Video**
 - **Text**
 - **Numbers**
-

Step 2: Feature Learning

Hidden layers automatically learn:

- **Patterns**
- **Shapes**
- **Relationships**
- **Important features**

Example:

- **Detecting eyes, nose, ears in images**
-

Step 3: Prediction

The output layer gives:

- **Classification**
- **Prediction**

- **Decision**

Example:

- **Cat or Dog**
 - **Spam or Not Spam**
-

Neural Network Formula

Each neuron calculates:

$$y = f(\sum w_i x_i + b)$$

Where:

- x_i = inputs
 - w_i = weights
 - b = bias
 - f = activation function
-

Important Components

1. Neurons

Basic processing units of the network.

2. Weights

Control importance of input values.

3. Bias

Helps shift predictions.

4. Activation Functions

Introduce non-linearity.

Common functions:

Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

ReLU

$$f(x) = \max(0, x)$$

Training Process

Deep learning models learn using:

1. Forward Propagation

Input moves forward through layers.

2. Error Calculation

Compare prediction with actual output.

3. Backpropagation

Error moves backward and updates weights.

Types of Deep Learning Networks

1. Artificial Neural Network (ANN)

Basic neural network.

Used for:

- **Classification**
 - **Prediction**
-

2. Convolutional Neural Network (CNN)

Used for image processing.

Applications:

- **Face recognition**
 - **Object detection**
-

3. Recurrent Neural Network (RNN)

Used for sequential data.

Applications:

- **Language translation**
 - **Speech recognition**
-

4. Long Short-Term Memory (LSTM)

Advanced RNN for remembering long sequences.

Applications:

- **Chatbots**
 - **Time-series prediction**
-

Applications of Deep Learning

1. Image Recognition

- **Face unlock**
- **Medical imaging**

2. Natural Language Processing (NLP)

- **ChatGPT**
- **Translation systems**

3. Speech Recognition

- **Siri**
- **Google Assistant**

4. Self-Driving Cars

- **Object detection**
- **Lane detection**

5. Recommendation Systems

- **Netflix**
- **YouTube**
- **Amazon**

Advantages

- **High accuracy**
- **Automatic feature extraction**
- **Handles complex data**
- **Excellent for images/audio/text**

Disadvantages

- **Requires huge data**
- **Needs powerful GPUs**
- **Training takes time**
- **Difficult to interpret**

Real-Life Example

Suppose you upload a photo to Instagram.

Deep learning can:

- **Detect faces**
- **Apply filters**
- **Recognize objects**
- **Suggest tags**

Machine Learning vs Deep Learning Example

Machine Learning

You manually tell the system:

- **“These are cat features.”**

Deep Learning

The system automatically learns:

- **ears**

- **eyes**
- **fur**
- **shape**

from large image datasets.